

Tibetan OCR Grid Search — Workflow Reference

Step-by-step guide for running, analyzing, and exporting OCR grid search results.

1. Before Starting a New Script Type Category

Each script type / quality combination (e.g. umeh_druma_high, umeh_drutsa_high) is treated as a separate category. The first file in each new category gets the full 1,728-combo grid. Subsequent files in the same category use the trimmed grid.

Back up the current summary.csv

Run this in the terminal before starting a new category:

```
cp ~/Documents/tibetan-ocr-app/grid_search_results/summary.csv  
~/Documents/tibetan-ocr-app/grid_search_results/summary_CATEGORYNAME_complete.csv
```

Replace CATEGORYNAME with e.g. umeh_druma_high. This preserves the previous category's results for standalone analysis.

2. Phase 1 — Full Grid on First Image of New Category

Open ocr_grid_search.ipynb and make these changes:

In the Configuration cell:

- Uncomment the TARGET for the first file in the new category. Comment out all other TARGET lines.
- Set MODELS_TO_TEST: use ["Woodblock", "Woodblock-Stacks", "Modern"] for Uchen; use ["Ume_Druma", "Ume_Petsuk", "Modern"] for Ume.

In the Parameters cell:

- Set the active selection to FULL_PARAMS:

```
PARAM_VALUES = FULL_PARAMS
```

In the Quick Test cell:

- Uncomment all three lines:

```
checkpoint = CheckpointManager(CHECKPOINT_FILE) checkpoint.reset() quick_results =  
run_grid_search(max_images=1, resume=False)
```

In the Full Run cell:

- Comment out:

```
# results = run_grid_search()
```

Run cells in order: setup cells first, then the Quick Test cell. This runs 1,728 combos on page 1 only (~5 hours).

■ Do NOT reset the checkpoint after this run — it marks page 1 as done.

3. Analyze Page 1 Results

Open `grid_search_analysis.ipynb` and run all analysis cells. Look for:

- Which parameters show no difference across values (candidates to trim to a single value)
- Which parameters show a clear winner (candidates to drop losers)
- Use composite score (`quality_score × total_tokens`) not just `quality_score` — a high score on 5 tokens is misleading

✓ Do NOT run the Export cell yet — wait until all pages in the category are complete.

4. Create Trimmed Parameter Set

Back in `ocr_grid_search.ipynb`, in the Parameters cell, add a new named param set based on your analysis. Follow the existing naming convention:

```
UMEH_DRUTSA_HIGH_PARAMS = { "line_mode": [...], "class_threshold": [...], "k_factor": [...], "bbox_tolerance": [...], "merge_lines": [...], "tps_threshold": [...], }
```

Add a comment above explaining what each choice is based on. Then update the active selection:

```
# PARAM_VALUES = FULL_PARAMS PARAM_VALUES = UMEH_DRUTSA_HIGH_PARAMS # or whatever the new name is
```

Also add the new param name to the ACTIVE SELECTION comment block so it's visible for future reference.

5. Phase 2 — Trimmed Run on Remaining Pages

The TARGET stays the same (same PDF). The checkpoint already has page 1 marked done — it will be skipped automatically.

In the Quick Test cell:

- Comment out all three lines:

```
# checkpoint = CheckpointManager(CHECKPOINT_FILE) # checkpoint.reset() #
quick_results = run_grid_search(max_images=1, resume=False)
```

In the Full Run cell:

- Uncomment:

```
results = run_grid_search()
```

Run all setup cells, then the Full Run cell. Pages 2+ will run with the trimmed grid.

■ **summary.csv appends — it will NOT overwrite page 1 results.**

6. Move to Next PDF in Same Category

If there are more PDFs in the same category (e.g. `umeh_dhernangdri` has 3 files):

- Change TARGET to the next PDF in the same category
- Keep PARAM_VALUES pointing to the same trimmed params (already tuned for this category)

- Keep the Full Run cell active, Quick Test cell commented out
 - The checkpoint auto-resets per image path so new PDFs start fresh
 - Run all setup cells then Full Run cell
- ✓ Do NOT reset the checkpoint manually — it tracks by full file path so different PDFs don't conflict.

7. Analysis and Export for Nyima

Once ALL PDFs in a category are done:

Run grid_search_analysis.ipynb:

- Run all analysis cells to review the full category results
- In the Export cell, uncomment the correct CATEGORY line:

```
CATEGORY = "umeh_druma_high" # match the category just completed
```

- Run the Export cell — it picks top 3 results per page by composite score, copies the original JPG, and saves clean OCR text to review_export/CATEGORY/pdf_name/

■ The Export cell reads ALL rows in summary.csv and groups by file_name. If summary.csv contains results from multiple categories, the export still works correctly because it separates by file_name — but be aware you're exporting everything, not just the current category. Use your backed-up per-category CSVs for clean per-category analysis.

8. Sync to Google Drive for Nyima

After the export cell completes:

```
rsync copy ~/Documents/tibetan-ocr-app/grid_search_results/review_export/
"gddrive:Tibetan OCR Gridsearch Project Folder/review_export" --progress
```

Use copy (not sync) to avoid deleting anything already on Drive.

9. Move to Next Category

Back up the summary.csv before starting a new category (see Section 1). Then:

- Change TARGET to first file of new category
- Change MODELS_TO_TEST if switching between Uchen and Umeh
- Set PARAM_VALUES = FULL_PARAMS
- Switch to Quick Test cell (uncomment), comment out Full Run cell
- Reset checkpoint: uncomment checkpoint.reset(), run it, re-comment it
- Run setup cells then Quick Test cell

Repeat from Section 3.

10. Quick Reference — What Each File Does

ocr_grid_search.ipynb — Runs OCR with parameter combinations. Outputs summary.csv and individual .txt result files.

grid_search_analysis.ipynb — Analyzes summary.csv. Last cell exports top results to review_export/ for Nyima.

summary.csv — Master results file. Appends across runs. Back up before each new category.

grid_search_results/review_export/ — Curated export for Nyima. Organized by category/pdf_name/. Synced to Google Drive via rclone.

grid_search_results/temp_images/ — Extracted page JPGs. Used by export cell to copy originals. Not for editing.

grid_search_results/_checkpoints/ — Tracks which images have been processed. Reset only when starting a new category fresh.

OCRModels/ — OCR model files. Downloaded once from GitHub release. Do not modify.

Models/Lines/ and Models/Layout/ — Line and layout detection models. Downloaded via git lfs pull. Do not modify.